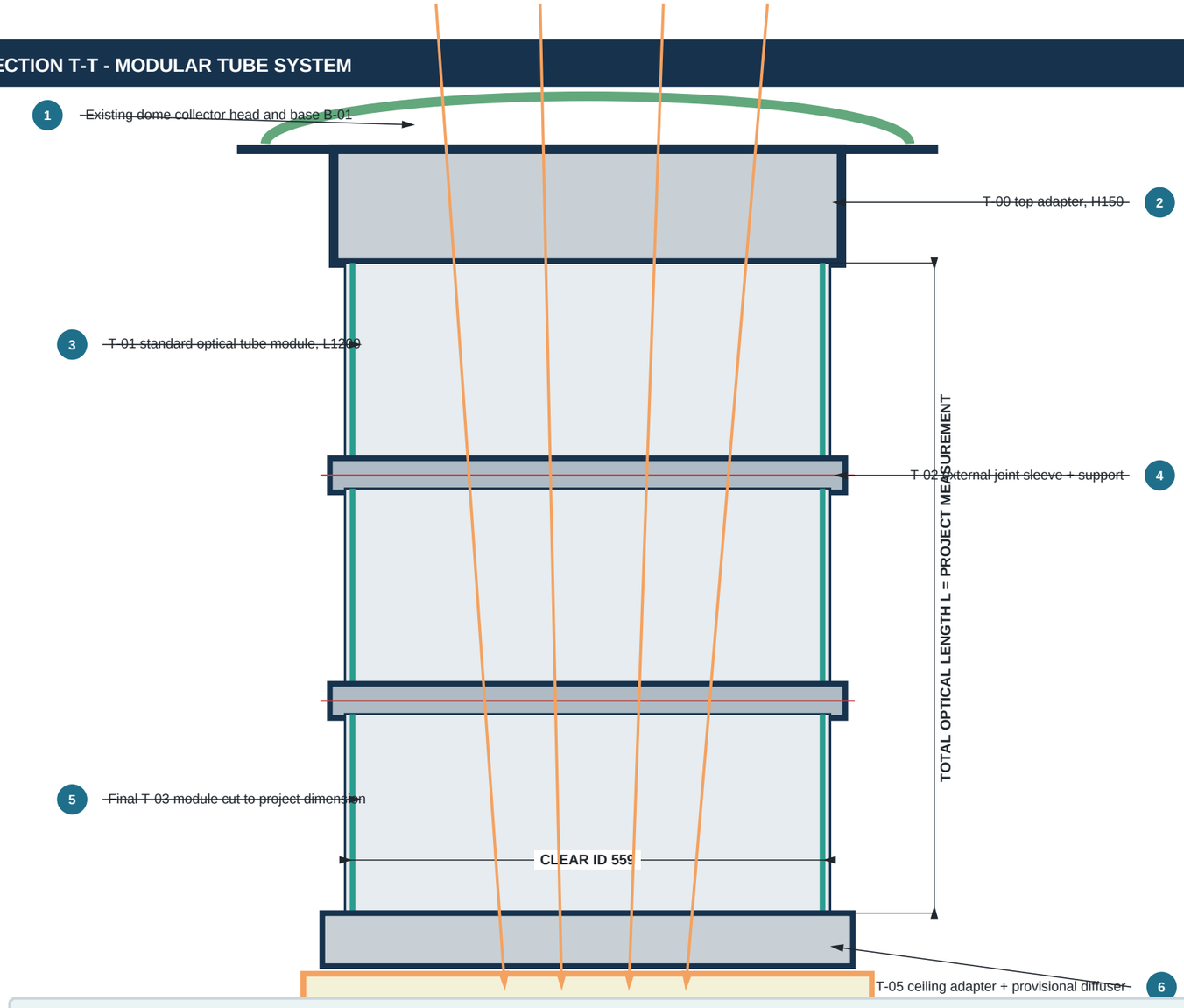
TUBULAR DAYLIGHT TUBE - SYSTEM GENERAL ARRANGEMENT

Modular straight optical tube matched to the dome package: clear bore DIA559, standard module length 1200

DESIGN BASIS: STRAIGHT VERTICAL PATH - TOTAL LENGTH L IS NOT YET KNOWN

Measure roof-collector throat to finished-ceiling interface before cutting the final module. Do not fabricate an offset or elbow from this package.

SECTION T-T - MODULAR TUBE SYSTEM



L = 1200 x N + Ltrim N = number of full modules; use 600 short module if required; preferred Ltrim = 300 to 1199

All module ends butt together: no insertion deduction. Interior step or axial gap at any joint: 1.0 maximum.

FIXED TUBE DESIGN BASIS

PARAMETER	PROTOTYPE VALUE
Clear optical bore	DIA559 continuous
Reflective surface	98% target; 95% minimum visible
Tube material	Enhanced-specular Al, 0.7 nominal
Standard / short length	1200 / 600
Study maximum L	12,000 pending photometry
Preferred routing	Straight and vertical

PROJECT DATA STILL REQUIRED

Measured collector-to-ceiling length L	TBD
Roof and ceiling centerline offset	TBD
Ceiling construction and diffuser type	TBD
Support structure / seismic category	TBD
Required insulation and vapor control	TBD
Fire / smoke rating for tube and insulation	TBD
Room illuminance and glare criteria	TBD

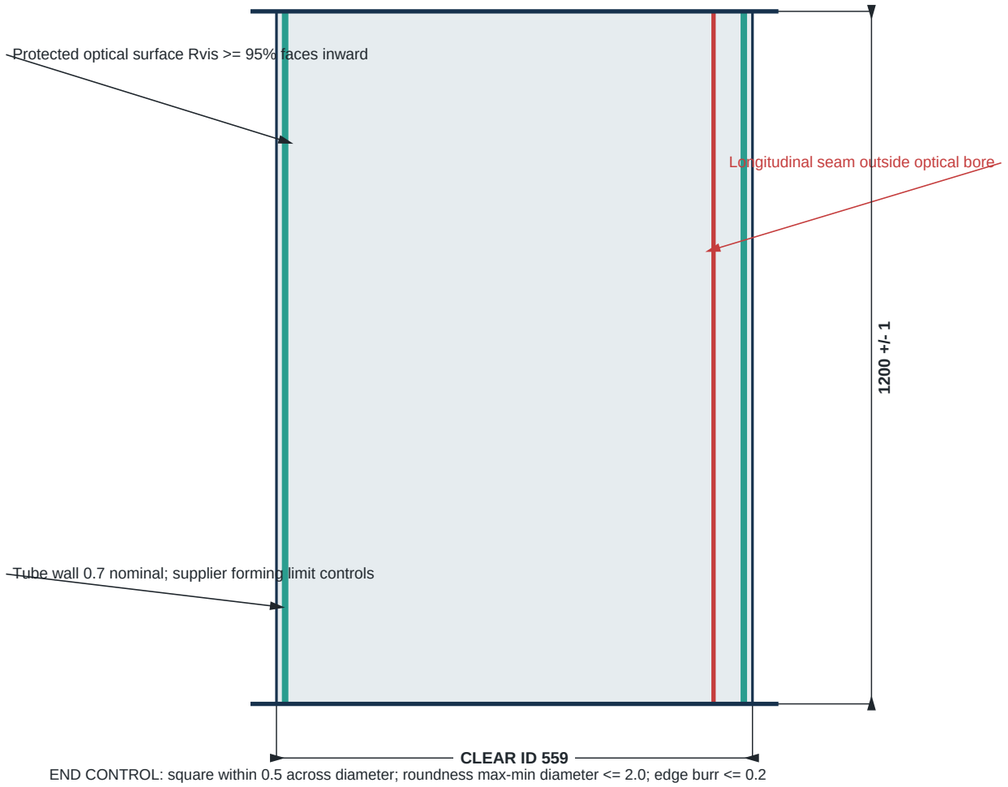
OPTICAL RULES

1. Keep the bore circular and every joint flush; steps scatter light and increase loss.
2. Use the highest practical reflectance. Loss compounds after every wall reflection.
3. Keep longitudinal seams aligned and smooth; place seams away from the dominant low-angle sun side.
4. Support the tube independently. Do not hang its weight from the dome base, roof membrane or ceiling grid.
5. Do not use flexible corrugated duct: its surface and geometry cause severe optical loss.

STRAIGHT OPTICAL TUBE MODULE - BLANK, ROLLING AND SEAM

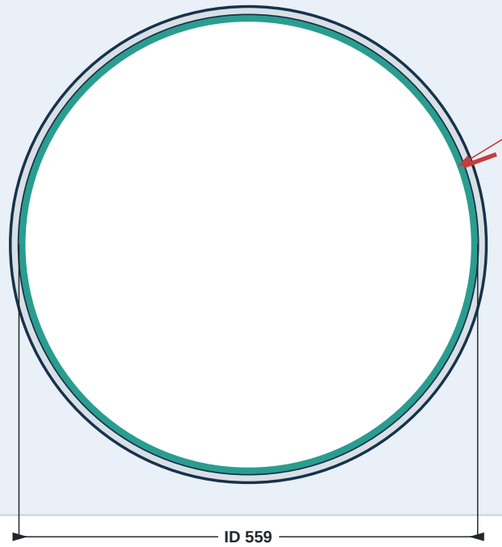
Part T-01: DIA559 clear bore x 1200 long; enhanced-specular aluminum, reflective face inward

PART T-01 - FINISHED MODULE



END CONTROL: square within 0.5 across diameter; roundness max-min diameter <= 2.0; edge burr <= 0.2

SECTION A-A - BORE AND LONGITUDINAL SEAM



ID at seam may reduce by 0.5 max

No rivet or screw may enter the bore

Reflective face scratch limit: 0.25 wide

Finished OD about 560.4 excluding seam

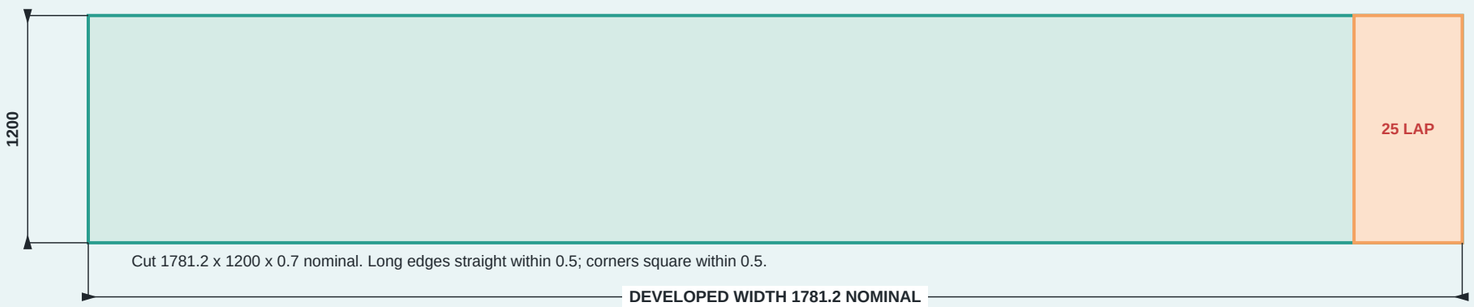
Module mass depends on supplied coil grade

Inspect ID at 0 / 45 / 90 / 135 deg

FLAT BLANK T-01-B - CUT BEFORE ROLLING

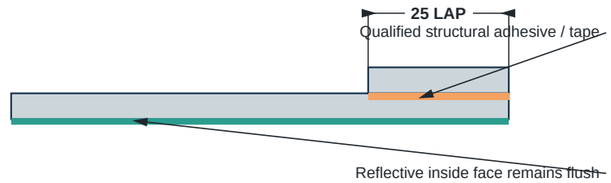
Calculation: pi x 559 clear circumference + 25 lap = 1781.2. Calibrate +/- 2 after the first mandrel trial.

Grain / coil direction runs parallel to tube axis. Keep protective film on optical face until seam closure.



Cut 1781.2 x 1200 x 0.7 nominal. Long edges straight within 0.5; corners square within 0.5.

DETAIL A - EXTERNAL BONDED LAP



1. Clean and prepare only by optical-sheet supplier method.
2. Roll over a DIA559 mandrel with optical face protected.
3. Bond 25 lap on the outside; use a full-length pressure bar.
4. Cure while held round. Do not spot-rivet through the bore.
5. Remove film, clean and inspect after all metalwork ends.
6. Make a seam coupon and peel/shear test before production.

ADHESIVE IS NOT YET SPECIFIED

Select after coil coating compatibility, temperature, humidity and fire review.

TUBE JOINT, TOP ADAPTER, SUPPORT AND INSULATION

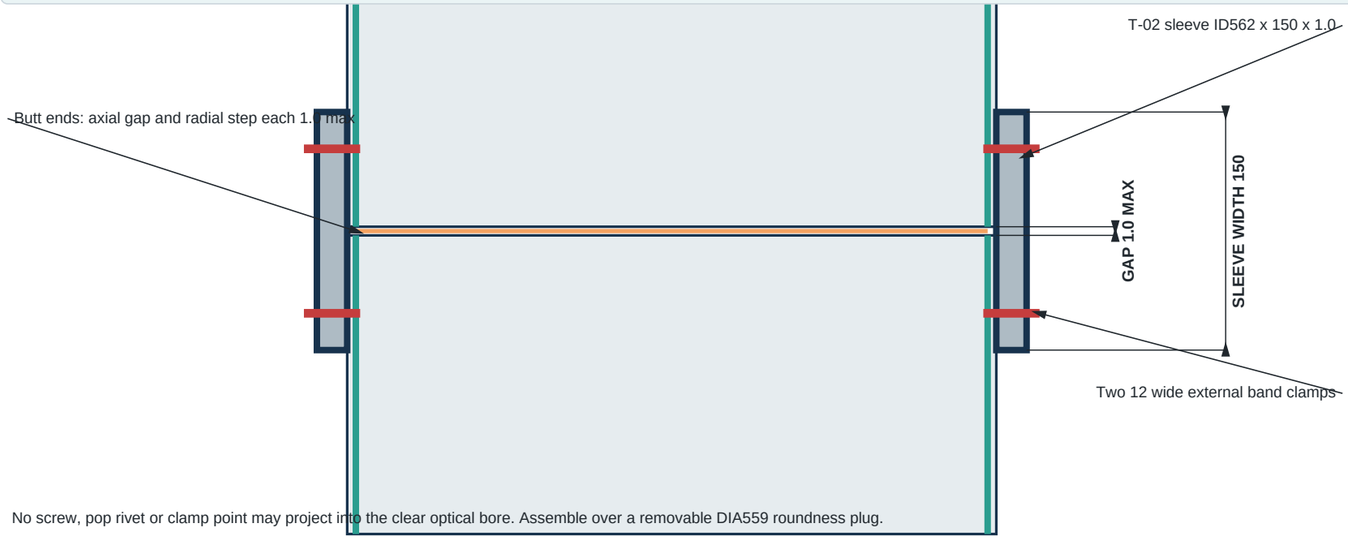
Non-penetrating butt joints preserve the DIA559 optical bore; supports remain independent of roof and ceiling finishes

UNITS: mm unless noted | SCALE: diagrammatic

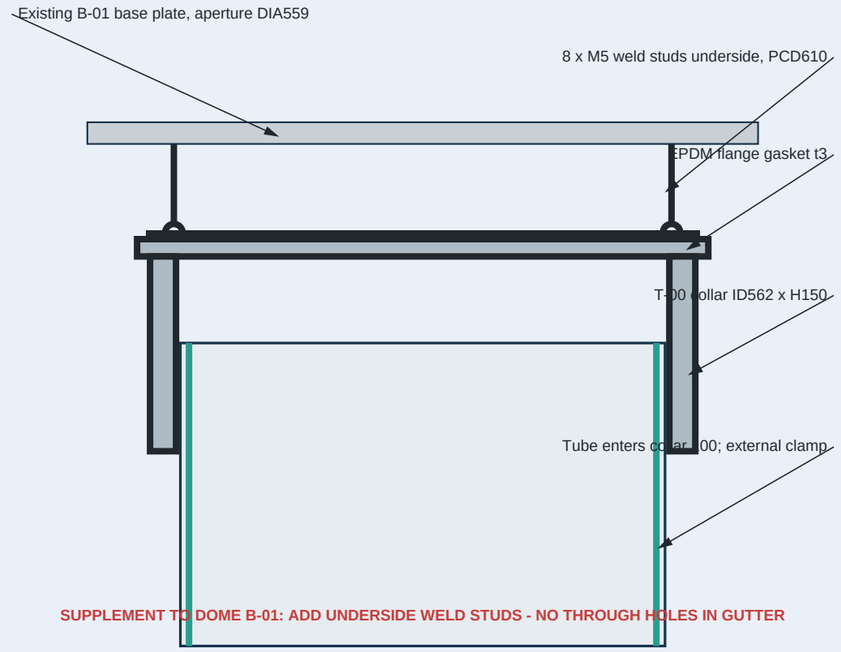
DETAIL J-1 - BUTT JOINT AND EXTERNAL SPLIT SLEEVE

T-02 BLANK: 1795 x 150 x 1.0 aluminum; 30 lap; roll to ID562

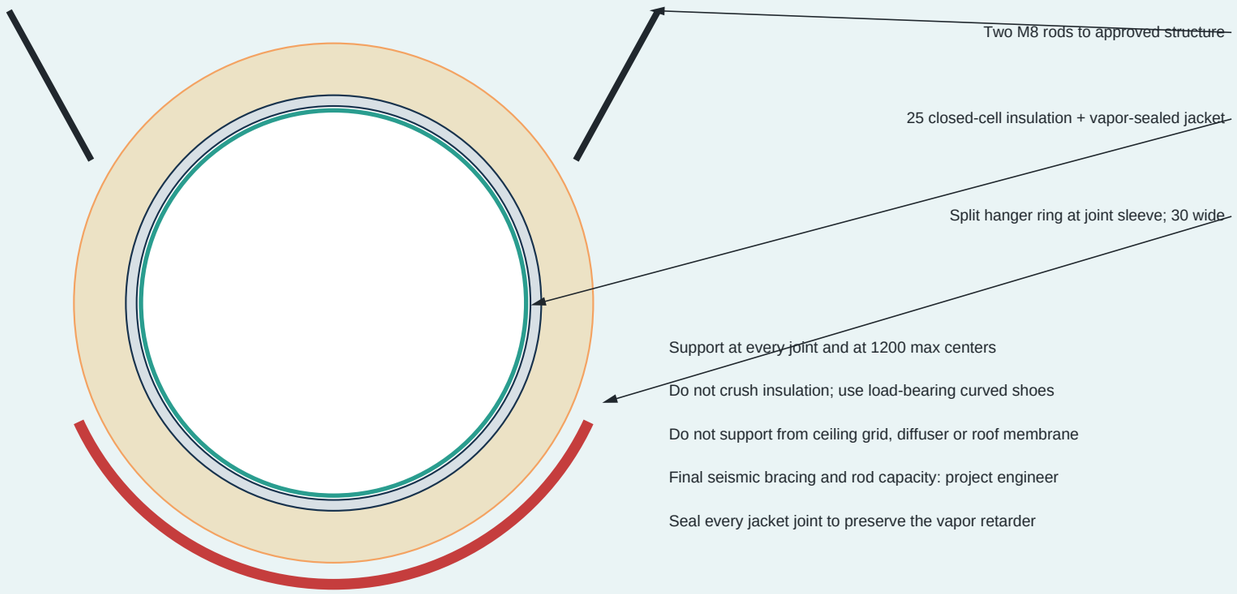
Developed width = pi x 562 + 30 = 1795.4 nominal. Slot the outer lap for clamp access; deburr all edges.



DETAIL J-2 - TOP ADAPTER TO DOME BASE B-01



DETAIL S-1 - INDEPENDENT SUPPORT AT EACH MODULE JOINT



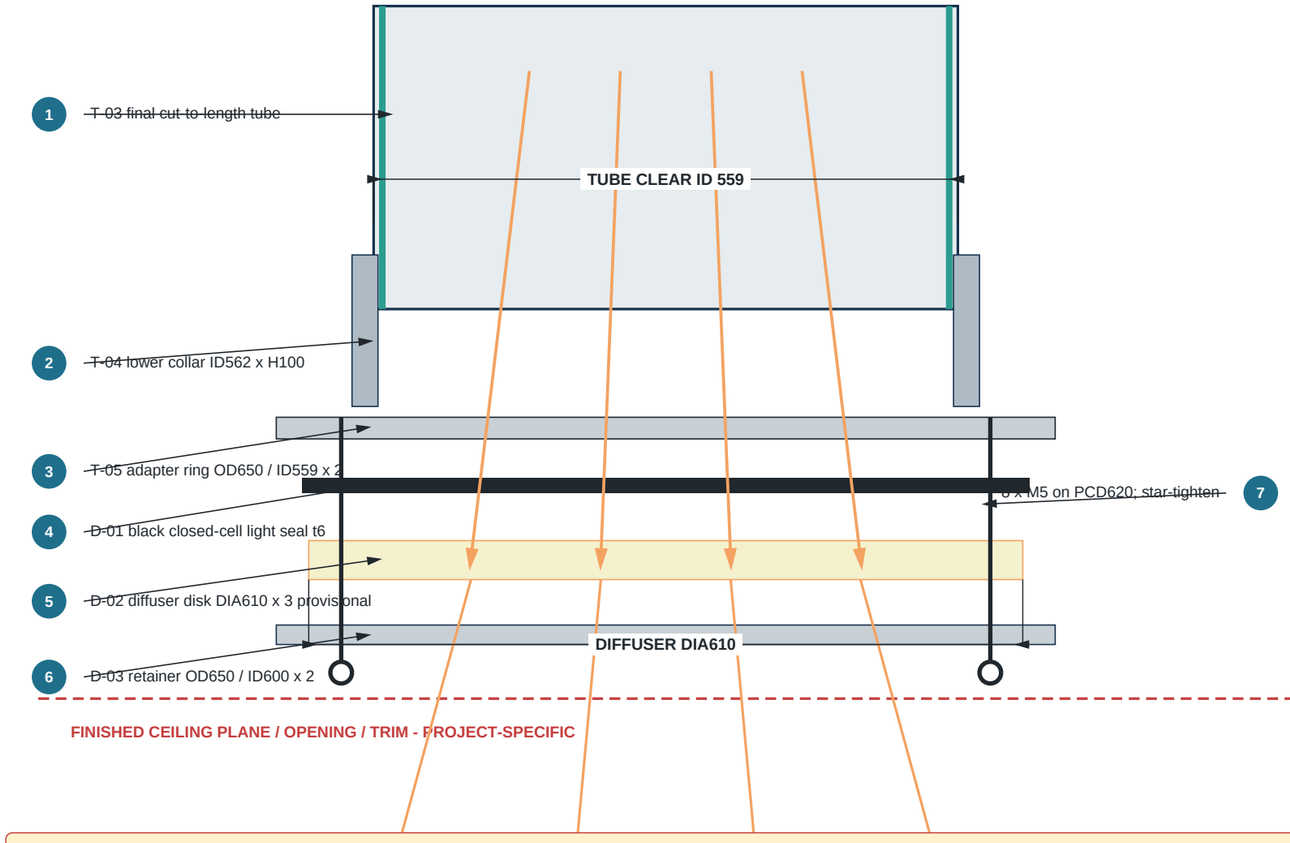
INSULATION / CONDENSATION CONTROL

1. Tube metal is highly conductive; uninsulated attic sections can condense interior moisture.
2. Prototype basis: 25 closed-cell insulation, continuous vapor-sealed jacket, approximate OD612.
3. Insulation type, thickness, flame spread, smoke development and thermal rating remain project-specific.
4. Seal the top adapter and lower terminal against air leakage, but keep the dome gutter drains open.
5. Provide inspection access at the first joint and at the lower terminal.
6. Do not place absorbent insulation inside the optical bore.

LOWER TERMINAL, DIFFUSER INTERFACE AND ROUTING CONTROL

Ceiling hardware is provisional until ceiling type, fire criteria and required light distribution are known

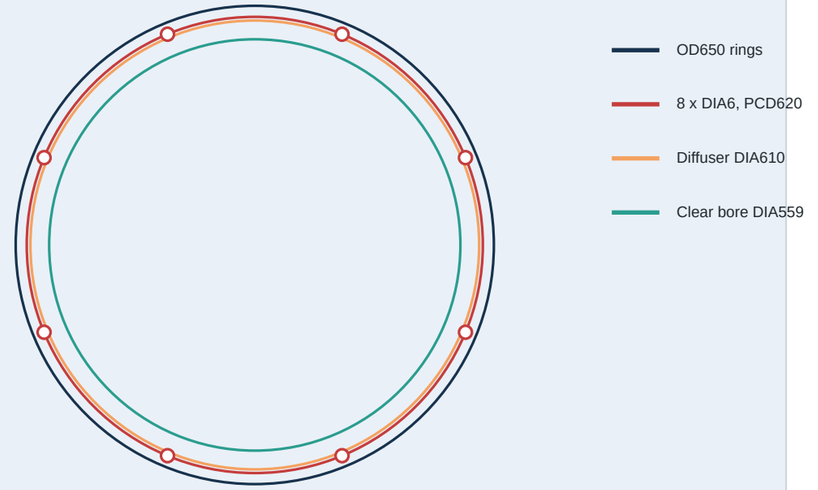
DETAIL D-1 - LOWER TERMINAL EXPLODED SECTION



DIFFUSER PERFORMANCE IS NOT RELEASED

Select optical pattern and transmission after room photometry and glare review. Confirm fire, impact, cleaning and ceiling-plenum requirements before manufacture.

TERMINAL RING - PLAN



OFFSET / ELBOW DECISION - HOLD

DEFAULT: NO ELBOW

1. Measure horizontal offset X between roof and diffuser centerlines.
2. If X <= 25, align straight modules and use terminal trim only.
3. If X > 25, stop fabrication and produce a project ray trace plus elbow development.
4. Do not force, ovalize or kink a straight reflective module to create an offset.
5. Every bend increases reflection count and reduces transmitted light.

CUT-LENGTH WORKSHEET

Measured optical length L = _____ Top adapter occupied length = 100 Lower collar occupied length = 75

Net straight tube length Ltube = L - 175 = _____

Number full 1200 modules N = floor(Ltube / 1200) = _____ Remaining cut length Ltrim = _____

If Ltrim < 300: replace one 1200 module with one 600 module, then recalculate. Minimum final cut module = 300.

Shop release: N x T-01 1200 modules + ____ x T-01S 600 module + one T-03 cut module _____ long.

DIFFUSER RELEASE INPUTS

Room use / target lux	TBD
Ceiling type and opening	TBD
Required transmission / haze	TBD
Glare criterion	TBD
Fire / smoke / impact rating	TBD

TUBE FABRICATION, OPTICAL CHECK AND RELEASE MATRIX

Manufacturing sequence, bill of materials and tests for the DIA559 modular daylight tube

A. MANUFACTURING AND ASSEMBLY SEQUENCE

01	**CONFIRM ROUTE**	Measure L and centerline offset; release straight layout only when X <= 25.
02	**RECEIVE COIL**	Verify supplier grade, visible reflectance certificate, coating direction and forming limits.
03	**CUT BLANKS**	T-01: 1781.2 x module length; T-02 sleeve: 1795 x 150; protect optical face.
04	**ROLL MODULES**	Use DIA559 mandrel; make external 25 lap; cure round; trim square.
05	**INSPECT**	ID, roundness, length, seam, scratches and reflectance evidence; reject defects.
06	**ADD ADAPTERS**	Install T-00 under dome base and lower T-04/T-05 terminal without bore projections.
07	**JOIN**	Butt modules over alignment plug; fit T-02 split sleeve and two external clamps.
08	**SUPPORT**	Hang at every joint / 1200 max centers from approved structure; add bracing as designed.
09	**INSULATE**	Apply continuous approved insulation and vapor jacket; seal every joint and penetration.
10	**TEST/RELEASE**	Check air, condensation, supports and photometry before enclosing the ceiling.

B. INSPECTION AND TEST MATRIX

ITEM	ACCEPTANCE	STATUS
Module length	1200 +/- 1; cut module per worksheet	OPEN
Clear bore	559 +/- 1; seam reduction <= 0.5	OPEN
Roundness	Max ID - min ID <= 2.0	OPEN
End squareness	0.5 across diameter	OPEN
Joint alignment	Gap and radial step each <= 1.0	OPEN
Optical surface	Rvis cert >=95%; target 98%; no major scratch	OPEN
Seam coupon	Pass qualified peel / shear criterion	HOLD
Air leakage	Project criterion; smoke pencil / pressure test	HOLD
Condensation	No interior wetting in design condition	HOLD
Support / seismic	Approved loads, spacing and bracing	HOLD
Fire / smoke	Tube, adhesive and insulation approved	HOLD
Photometric	Room lux and glare criteria achieved	HOLD
Access / cleaning	Top, first joint and terminal accessible	OPEN

C. APPROXIMATE REFLECTION LOSS - STRAIGHT TUBE

For a simplified central ray: number of wall hits N approximately equals L x tan(theta) / D; surface throughput approximately rho^N.

Example below: L = 6000, D = 559. This excludes dome, joint, dirt and diffuser losses.

RAY ANGLE TO AXIS	N APPROX	AT rho=0.95	AT rho=0.98
15 deg	3	86%	94%
30 deg	6	74%	89%
45 deg	11	57%	80%

DESIGN CONSEQUENCE: 98% reflector material and a straight, clean bore materially improve delivered light.

Use measured photometry for release. This simplified relation is for design comparison, not a product rating.

D. BILL OF MATERIALS AND TECHNICAL BASIS

PART	QTY	DESCRIPTION
T-00	1	Top collar ID562 x H150 + OD640 flange
T-01	N	Optical tube ID559 x 1200 x 0.7 nominal
T-01S	opt	Optical tube ID559 x 600 x 0.7 nominal
T-02	N	External split sleeve ID562 x 150 x 1.0
T-03	1	Final optical tube, cut length 300-1199
T-04	1	Lower collar ID562 x H100
T-05	1	Terminal adapter OD650 / ID559 x 2
D-01	1	Closed-cell light seal t6
D-02	1	Diffuser DIA610 x 3 - PROVISIONAL
D-03	1	Retainer OD650 / ID600 x 2
S-01	N+1	Split support ring + two M8 hanger rods
I-01	L	Approved insulation / vapor jacket - TBD

Basis: Alanod daylight reflector surfaces - visible reflection up to 98%; coil and cut-sheet formats.

Basis: VELUX commercial 22 in tubular daylight devices - high-reflectance rigid tube and modular connections.

Basis: FGIA NAFS - TDDCC and TDDOC classifications; adopted edition and performance criteria are project-specific.

This is a generic prototype design and does not reproduce a proprietary connector, elbow or diffuser.